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Speculative Decoding

Problem: an LLM generates one token per forward pass, and each pass of a large model is limited by **memory bandwidth** (all weights read for one token), leaving GPU compute idle. **Idea:** a small, fast **draft model** q guesses the next k tokens, and the large **target model** p checks all k guesses in **one** forward pass, which costs about the same as generating one token. Correct guesses are kept for free.

Algorithm (one round):

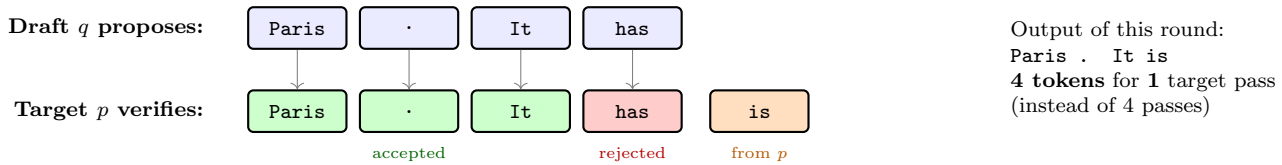
(1) **Draft:** the small model generates k tokens autoregressively: $\tilde{x}_1, \ldots, \tilde{x}_k$ (cheap).

(2) **Verify:** the target model scores the prefix plus all k draft tokens in parallel, getting $p(\cdot)$ at every position.

(3) **Accept / reject** each draft token in order, with probability $\min\Big(1, \frac{p(\tilde{x}_i)}{q(\tilde{x}_i)}\Big)$. At the first rejection, sample a replacement from $\text{norm}\big(\max(0,\, p - q)\big)$ and stop. If all k are accepted, the target pass also yields one **bonus** token.

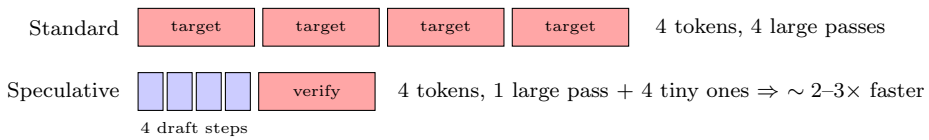
This rule keeps **exactly the same output distribution** as the target alone: faster, no quality loss (greedy: accept while draft = target's argmax).

Example: prompt `"The capital of France is"`, draft length $k = 4$.



The draft got `Paris`, `.` and `It` right (easy, predictable tokens), so they are kept; `has` was unlikely under the target, so it is replaced by the target's own choice `is`. The next round starts from `"... Paris. It is"`.

Timeline: standard vs speculative



Expected speedup: with acceptance rate α, the expected tokens per target pass are $\frac{1-\alpha^{k+1}}{1-\alpha}$; for $\alpha = 0.8$, $k = 4$: $\frac{1-0.8^5}{0.2} \approx 3.4$. The draft must be **much cheaper** yet **agree often** with the target (same family and tokenizer, e.g. 1B draft for 70B target). Variants without a separate draft: **Medusa**, **EAGLE**, **prompt lookup**. Supported in vLLM, TensorRT-LLM, SGLang and Hugging Face.